Exact Rational Integration over Constrained Simplices

Symmetric Polynomial Bases, Marginal Geometry, and Lean 4 Verification

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Abstract

We present a formalized collection of integration identities for polynomial weights on simplexes constrained by coordinate thresholds and cardinality-dependent caps. The development includes exact simplex moments, a rational polynomial subspace of dimension 846, cap-parametric formulas for monomial integrals, rational Gram-entry representations, fibre decompositions, and two-aggregate Dirichlet integration. Exact polygon-cell formulas cover all marginal endpoint branches. Kernel-checked rational certificates establish both the constant entry $J_T[1,1]$ and the nonconstant power-sum entry $J_T[P_2,1]$ under Table 3 parameters. A bilinear marginal form and coordinate-placement accounting underpin the new entry. The Lean 4 development passed all 19 Comparator suites on the VPS. These are independent integration results motivated by a bounded-prime-gap argument; the next target is $J_T[P_2,P_2]$ and eventually variational positivity.

1 Introduction and scope

Exact integration on constrained regions is useful where a variational inequality must ultimately be checked using rational arithmetic. A conventional simplex has a closed monomial moment formula; the present support also depends on which coordinates cross a threshold and on a cap for the sum of those coordinates. This introduces a finite stratification, but preserves a route to exact, rational expressions.

We study the mathematical constructions in the public Lean repository:

`github.com/thinkthoughts/lean-bgp212`

The motivating bounded-prime-gap argument originates with Axiom Math [1]; the statements below are an independent formalization of specified components and do not amount to a complete formal proof of the unconditional prime-gap bound. References to Lean declarations are included to delimit precisely what has been checked. The short arguments given here are *proof outlines*, not substitutes for the complete Lean proof terms.

2 Polynomial moments on a simplex

Definition 2.1. *For $m \in \mathbb{N}$ and $R \in \mathbb{R}$, define*

$$\Delta_m(R) = \left\{ x \in \mathbb{R}^m : x_i \geq 0 \text{ for all } i, \quad \sum_{i=1}^{m} x_i \leq R \right\}.$$

Theorem 2.2 (Simplex monomial moments)**.** *For $m \in \mathbb{N}$, $R \geq 0$, and $e \in \mathbb{N}^m$,*

$$\int_{\Delta_m(R)} \prod_{i=1}^{m} x_i^{e_i}\, dx = \frac{R^{m+\sum_i e_i} \prod_i (e_i!)}{(m+\sum_i e_i)!}.$$

Here the dimension-zero case uses the corresponding Lean integral convention.

Lean: `simplex_integral`; definitions `simplex`, `momentR`.
Source: `Solution/SimplexMoment.lean:160`.

Proof outline. Induct on dimension and use Fubini to isolate a coordinate. The inner integral is a Beta integral; the factorial expression then simplifies by induction. The formal development additionally treats negative and zero radii, simplex volume, and rational-polynomial integrals (`poly_integral`).

3 An indexed rational polynomial space

Consider rational polynomials in 45 coordinates. Write P_j for the jth power sum, P_λ for the product over parts of a multiset λ, and set

$$G_{a,\lambda} = (L - P_1)^a P_\lambda, \qquad \lambda \text{ has positive even parts}, \quad a + |\lambda| \le 21.$$

The index type $B_{21}\mathrm{Idx}$ enumerates the admissible pairs.

Theorem 3.1 (Independence and rank)**.** *The indexed family $G_{a,\lambda}$ is linearly independent over $\mathbb{Q}$. Its rational span B_{21} has*

$$\dim_{\mathbb{Q}} B_{21} = 846.$$

Lean: `genIdx_linearIndependent`, `finrank_B21` (`Solution/Variational.lean:398,410`).

Proof outline. The indexing is injective; distinct indexed monomials are linearly independent. Injective polynomial transformations preserve this independence. A separate count establishes $|B_{21}\mathrm{Idx}| = 846$, and the span-rank identity gives the result. The theorem concerns this designated subspace, not the full polynomial ring.

4 Cap-constrained supports and general integrals

Let $N \in \mathbb{N}$, $g \ge 0$, $L \in \mathbb{R}$, and $C : \mathbb{N} \to \mathbb{R}$. For $x \in \mathbb{R}^N$ write

$$R_g(x) = \{i : x_i > g\}, \qquad s_g(x) = |R_g(x)|, \qquad \rho_g(x) = \sum_{i \in R_g(x)} x_i.$$

Define the capped simplex support

$$T(N, g, L, C) = \left\{ x \in \mathbb{R}^N : x_i \ge 0, \quad \sum_i x_i \le L, \quad \rho_g(x) \le C(s_g(x)) \right\}. \tag{1}$$

Theorem 4.1 (General monomial integral)**.** *For $N \in \mathbb{N}$, $g \ge 0$, $L \in \mathbb{R}$, $C : \mathbb{N} \to \mathbb{R}$, $e \in \mathbb{N}^N$, and $p \in \mathbb{N}$,*

$$\int_{T(N,g,L,C)} \left(L - \sum_i x_i \right)^p \prod_i x_i^{e_i} \, dx = \mathrm{gramGen}(N, g, L, C, e, p). \tag{2}$$

The right-hand side is the explicit, finite expression defined in Lean.

Lean: `integral_supportT_gen`; `gramGen`.
Sources: `Solution/GramGeneral.lean:344`;
`BGP212/GramGeneral.lean`.

Proof outline. Stratify by the rough-coordinate set, apply inclusion–exclusion to threshold conditions, and translate the resulting pieces to weighted simplexes. Use Theorem 2.2 and Fubini to evaluate the polynomial weights. Placement-counting identities `sig_insert`, `sig_union`, and `sig_card` reduce the sums to the finite closed form. The reduction `gramGen_extend` transports calculation to the coordinates having nonzero exponents. This is an algebraic efficiency result, not a claim that all entries are computationally cheap to certify.

5 Rational Gram representations

Specialize to the fixed rational cap values and support parameters of Table 3 in [1], with $N = 45$. For indices $i, j \in B_{21}\mathrm{Idx}$ define

$$I_T(i,j) = \int_T G_i(x) G_j(x)\, dx.$$

Theorem 5.1 (Explicit rational representation)**.** *For every $i, j \in B_{21}\mathrm{Idx}$, $I_T(i,j)$ equals the real embedding of a finite rational sum indexed by placements φ of the combined power-sum parts:*

$$I_T(i,j) = \left(\sum_{\varphi} \mathrm{gramGen}_{\mathbb{Q}}(45, g, L, C, e_\varphi, a_i + a_j) \right)_{\mathbb{R}}.$$

Here e_φ is the exponent vector obtained from the placement, exactly as specified by ***`expo`*** *in the Lean source.*

Lean: `gram_entry_B21_rat` (`Solution/GramGeneral.lean:522`); `gram_entry_gen_list` and `gram_entry_example`.

The theorem gives each entry an explicit rational representation; it does not supply an evaluated and certified 846×846 matrix. One worked entry, corresponding to $(L - P_1)P_2^2$, has its value certified using a collection of small exact rational lemmas. This separation between an algebraic identity and its numerical certificate is important to reproducibility.

6 Marginal-fibre geometry

Let $\widehat{x} \in \mathbb{R}^n$ and append $u \in \mathbb{R}$ as the last coordinate. Define $S = \sum_i \widehat{x}_i$, $s = s_g(\widehat{x})$, $\rho_1 = \rho_g(\widehat{x})$, and

$$\widehat{T} = \{\widehat{x} \geq 0 : S \leq \tau,\ \rho_1 \leq C(s)\}, \quad U(\widehat{x}) = \min\{L - S, \max(g, C(s+1) - \rho_1)\}.$$

Theorem 6.1 (Fibre equivalence)**.** *Suppose $g \geq 0$, $\tau \leq L$, and $\widehat{x} \in \widehat{T}$. Then for every $u \in \mathbb{R}$,*

$$(\widehat{x}, u) \in T(n+1, g, L, C) \quad \Longleftrightarrow \quad 0 \leq u \leq U(\widehat{x}).$$

Lean: `fibre_equiv` (`Solution/Fibre.lean:115`).

Proof outline. Appending u leaves the rough set unchanged for $u \leq g$, and increments both rough count and rough mass for $u > g$. Combining the resulting cap inequality with $S + u \leq L$ identifies the minimum–maximum endpoint exactly. The empty-fibre theorem for cap-violating outer coordinates additionally needs $C(m+1) \leq C(m) + g$ for $m \geq 1$, and $C(0) \geq 0$. No constraint on the step $m = 0$ is assumed; the relevant Table 3 facts are checked in Lean.

Theorem 6.2 (Marginal identity)**.** *Assume $g \geq 0$, $\tau \leq L$, $C(0) \geq 0$, and $C(m+1) \leq C(m) + g$ for every $m \geq 1$. For continuous P,*

$$\mathrm{marginalJ}(n, g, L, \tau, C, P) = J_T(n, g, L, \tau, C, P).$$

The right-hand functional integrates the squared fibre integral over $\widehat{T}$.

Lean: `marginal_agrees` (`Solution/Fibre.lean:217`).

Proposition 6.3 (Polynomial inner integral)**.** *For arbitrary $U, M \in \mathbb{R}$ and $p, t \in \mathbb{N}$,*

$$\int_0^U (M-u)^p u^t \, du = \sum_{j=0}^{p} \binom{p}{j} M^{p-j} (-1)^j \frac{U^{t+j+1}}{t+j+1}.$$

Lean: `inner_integral` (`Solution/Fibre.lean:240`). The equality follows by expanding the binomial and integrating each power. It holds for arbitrary real U under the oriented interval-integral convention.

7 Two-aggregate Dirichlet integration

Let I and J be disjoint, nonempty finite coordinate groups, with exponent vectors $a \in \mathbb{N}^I$ and $b \in \mathbb{N}^J$. For a nonempty finite group G and an exponent vector e, define

$$\alpha_G(e) = |G| + \sum_{i \in G} e_i - 1, \qquad A_G(e) = \frac{\prod_{i \in G}(e_i!)}{\alpha_G(e)!}.$$

Write $\Delta_{I \sqcup J}(M)$ for the nonnegative simplex with total coordinate sum at most M, and

$$\Delta_2(M) = \{(u_1, u_2) \in \mathbb{R}^2_{\geq 0} : u_1 + u_2 \leq M\}.$$

Theorem 7.1 (Two-aggregate Dirichlet integration)**.** *Suppose I and J are nonempty, and $h : \mathbb{R}^2 \to \mathbb{R}$ is measurable and bounded on $\Delta_2(M)$. With $u_1 = \sum_{i \in I} y_i$ and $u_2 = \sum_{j \in J} y_j$, the weighted simplex integral satisfies*

$$\begin{aligned} &\int_{\Delta_{I \sqcup J}(M)} \left(\prod_{i \in I} y_i^{a_i} \right) \left(\prod_{j \in J} y_j^{b_j} \right) h(u_1, u_2) \, dy \\ &\qquad = A_I(a) A_J(b) \int_{\Delta_2(M)} u_1^{\alpha_I(a)} u_2^{\alpha_J(b)} h(u_1, u_2) \, du_1 \, du_2. \end{aligned} \tag{3}$$

The formalization also gives the version on $\operatorname{Fin} N$ *for a rough coordinate set and its complement.*

Lean: `two_group`, `two_group_fin`, `one_group`; `Solution/Aggregate.lean`. The theorem is proved for the corresponding measurable, bounded integrands; separate theorems handle empty groups and other boundary cases.

Proof outline. Use Fubini and a sum-coordinate shear to isolate the total in one group; apply the weighted simplex moment calculation to the corresponding slices. Repeat for the second group. The factors A_I, A_J record the Dirichlet normalizations; in particular the -1 in each aggregate exponent is essential. The polynomial cases `tri_poly` and `two_group_poly` yield explicit rational expressions, and `two_group_const` recovers the simplex moment formula when $h = 1$.

Proposition 7.2 (A cut-triangle certificate)**.** *With $y = (y_0, y_1, y_2)$, the exact value is*

$$\int_{\substack{y_i \geq 0,\ y_0 + y_1 + y_2 \leq 1 \\ y_0 \leq 1/2}} y_0 y_2^2 \, dy = \frac{19}{7680}.$$

Lean: `tri_cut`, `example_value` (`Solution/Aggregate.lean`). The certificate reduces this integral to $\frac{1}{3} \int_{\Delta_2(1),\, u_1 \leq 1/2} u_1 u_2^3 \, du_1 du_2$; the value has also been cross-checked by direct rational integration.

Connection to marginal geometry. On the translated stratum $x = y + g\mathbf{1}_{R\cup K}$, define $u_1 = \sum_{i\in R} y_i$ and $u_2 = \sum_{i\notin R} y_i$. The proved correspondence `stratumEndpoint_shift` expresses the fibre endpoint as

$$U = \min\big(L - (r+k)g - u_1 - u_2,\ \max\big(g, C(r+1) - rg - u_1\big)\big)\,, \qquad r = |R|,\ k = |K|.$$

The cap index belongs to the stratum R. Substituting the translated point's rough set $R \cup K$ instead changes the endpoint; the Table 3 control records the distinct values 587/1250 and 461/1250. The cap-branch polygon-cell integration is established in Section 8; all endpoint branches and the constant marginal Gram entry are treated in Section 9; the power-sum entry follows in Section 10.

8 Polygon-cell integration and the cap branch

The two-aggregate reduction produces planar regions bounded by coordinate thresholds and a total-sum constraint. For real A, B, D define the clipped cell

$$\mathcal{B}(A, B, D) = \{(u_1, u_2) \in \mathbb{R}^2 : 0 \le u_1 \le A,\ 0 \le u_2 \le B,\ u_1 + u_2 \le D\}.$$

Theorem 8.1 (Exact polygon-cell moments)**.** *For $\alpha, \beta \in \mathbb{N}$, the Lebesgue integral*

$$\int_{\mathcal{B}(A,B,D)} u_1^\alpha u_2^\beta \, du_1\, du_2 = \mathrm{boxVal}(A, B, D, \alpha, \beta)$$

where boxVal *is a finite closed form using polynomial antiderivatives. For rational A, B, D its value is rational. The same construction gives rational integrals for arbitrary rational polynomials in two variables. Degenerate and empty cells are treated separately.*

Lean: `box_monomial`, `boxVal_cast`, `box_poly`, `boxVal_empty`, `boxVal_nonpos`, `boxVal_rect`; `Solution/CapCell.lean`.

Proof outline. Integrate first in u_2 with upper endpoint $\min(B, D - u_1)$ and split the u_1 interval at $D - B$. The resulting one-variable powers have exact polynomial antiderivatives. In particular, when $A + B \le D$ the diagonal constraint is redundant and the cell is a rectangle; the formula also covers diagonally truncated cells.

Theorem 8.2 (Cap-branch endpoint and piece integral)**.** *On a translated stratum $x = y + g\mathbf{1}_{R\cup K}$, with $r = |R|$, $k = |K|$ and aggregates u_1, u_2 as in Section 7, the cap branch of the fibre endpoint is*

$$U = C(r+1) - rg - u_1,$$

subject to its vertical and horizontal branch constraints. Combined with the piece constraints, the aggregate domain is a clipped cell $\mathcal{B}(A, B, D)$ with parameters determined by g, L, τ, C, r, k. The weighted piece integrals

$$\int_{\text{piece}} y^t\, \mathbf{1}_{\text{cap}} U^m \, dy$$

therefore equal an explicit expression obtained from the two-aggregate Dirichlet constants and rational cell integral formulas.

Lean: `endpointAgg_on_capCell`, `cap_branch_piece`, `cap_branch_r0`, `cap_inner`; `Solution/CapCell.lean`. The formalization proves the branch classification and corresponding endpoint equalities. For Table 3 the τ constraint is redundant on the cap branch: $\tau - (L - g) = 1/625 > 0$, and the cap cells become rectangles (`tau_ge_table3`, `capCell_rect_table3`).

Proposition 8.3 (A Table 3 cap-piece certificate)**.** *For the specified 44-coordinate piece* $(R, K) = (\{0\}, \varnothing)$*, the cap-branch contribution to the integral defining* $J_T(1)$ *is*

$$\frac{(L - C(2))^{43}((C(2) - g)^3 - g^3)}{3 \cdot 43!} \approx 1.0464 \times 10^{-70}.$$

Lean: `capPieceT3_value` (`Solution/CapCell.lean`). This is *one piece* of the inclusion–exclusion decomposition, not the complete value of $J_T(1)$. The exact rational certificate is kernel-checked. Polygon clipping and lower-dimensional integration provide independent numerical controls; the latter is an approximation, not part of the kernel proof. The remaining branches and their assembly are treated in Section 9.

9 All endpoint branches and the first marginal Gram entry

The translated marginal endpoint is the minimum of the remaining total mass and the maximum of a cap expression and g:

$$U = \min(L - (r + k)g - u_1 - u_2,\ \max(g, C(r + 1) - rg - u_1)).$$

Its switching lines partition each two-aggregate piece into four cells: the cap branch, the remaining-mass branch on the cap side, the constant g branch, and the remaining-mass branch on the g side. Each cell is represented by signed differences of the clipped polygon cells $\mathcal{B}(A, B, D)$ in Section 8.

Theorem 9.1 (Exact branch-cell and piece integrals)**.** *For nonnegative integer exponents, the weighted integral on each branch cell of a polynomial block*

$$u_1^{\alpha} u_2^{\beta} (p - u_1)^{m_1} (L' - u_1 - u_2)^{m_2}$$

reduces to a finite combination of exact clipped-cell moments. The corresponding integrals on translated rough-set strata admit closed forms, including the stratum with no rough coordinates. For integrands depending on the stratum, inclusion–exclusion and cardinality counting assemble these pieces into the marginal functional.

Lean: `box_twoPow`, `cells_integral`, `piece_block`, `piece_r0`, `indicator_supportT_strat`, `count_by_card`; `Solution/MarginalEntry.lean`.

Theorem 9.2 (Rational reduction of the constant-generator marginal)**.** *For* $N \geq 1$ *and parameters satisfying* $0 \leq g \leq C(1)$ *and* $C(N) < Ng$*, the squared-fibre marginal for the constant polynomial is expressed by the explicit rational-piece expression* jtSum*:*

$$J_T(1) = (N + 1) \int_{\widehat{T}} U(\widehat{x})^2 \, d\widehat{x} = \mathrm{jtSum}(N, g, L, \tau, C, 2, 0).$$

Lean: `jT_one_eq`, `jT_one_pieces`, `piece_eq_pieceJ`; `Solution/MarginalEntry.lean`. The reduction uses binomial counting for this symmetric integrand; entries with power-sum factors require additional expansion and placement accounting.

Proposition 9.3 (Table 3 marginal Gram certificate)**.** *Under the Table 3 parameters, for the generator* $G_{0,\varnothing} = 1$*,*

$$J_T[G_{0,\varnothing}, G_{0,\varnothing}] = 45 \int_{\widehat{T}} U(\widehat{x})^2 \, d\widehat{x} \approx 1.1193487646397893 \times 10^{-55}.$$

The exact value is the ratio of these integers, obtained by concatenating each integer's digit lines (no separators):

Numerator

19640530134259849383021060391154484358281020458705508076962340943624589435319293948822187291213238184493255346265105306516171857869871603814187

Denominator

*17546390146399319350401195543355143747760423948217811338346974871720647046191565721073857275769114494323730468750000000000

Lean: `p2EntryVal_table3`, `jT_P2_one_table3`, `jT_one_P2_table3`.
Sources: `Solution/PowerSumEntry.lean`, `Solution/PowerSumEntryCert.lean`.

The exact Table 3 evaluation uses 133 cardinality-class pieces with two certificate lemmas per piece (266 lemmas total); the remaining classes are eliminated by `cond_tail` and `cond_high`. The generator `specs/checks/power_sum_entry_v2g_emit.py` reproduces the Lean certificate byte-for-byte on the VPS. Six independent Lean-file controls have their expected outcomes: two accepted positive controls and four rejected negative controls. Numerical cross-checks include a low-dimensional grid and Monte Carlo; these support the arithmetic independently of the exact kernel proof. The next polynomial cross-entry is $J_T[P_2, P_2]$, requiring two distinguished-coordinate placements and a diagonal case.

11 Verification, limitations, and further mathematics

The project uses Lean `v4.34.0-rc2` and a pinned Mathlib revision `85e3a25`. At the V2g baseline, all 19 Comparator suites were independently accepted on the VPS. Four positive V2d controls were accepted; four intentionally incorrect controls were rejected with unsolved goals. The V2d proof commit is `52fd36a`; its published verification record is commit `90055db`. For V2e-A, five positive controls were accepted and four negative controls were rejected. The V2e-A proof commit is `01644e6`, with VPS verification/provenance commit `38606c0`. For V2e-B, five positive controls were accepted and five deliberately incorrect controls rejected; the V2e-B proof commit is `6a21780`, with verification/provenance commit `25a7376`. For V2f, four positive Lean-file controls were accepted and five negative controls rejected. Its proof commit is `8c50fb6`, with VPS provenance commit `da0c3fd`. For V2g, two positive controls were accepted and four negative controls rejected; its 266 certificate lemmas regenerated byte-identically on the VPS. The V2g proof commit is `ad70abb`, with verification/provenance commit `12f8b16`. The reported kernel axioms for these developments are `propext`, `Quot.sound`, and `Classical.choice`. Results and verification scripts are available in `specs/results`.

Further work begins with $J_T[P_2, P_2]$ and extends to general power-sum marginal Gram entries and their exact rational evaluation, optimized polynomial coefficients, exact variational positivity certificates, and remaining analytic hypotheses. The existing Lean development formalizes the *conditional* implication $\mathrm{DHL}[45, 2] \Rightarrow H_1 \leq 212$; this manuscript does not claim an unconditional formal proof of $H_1 \leq 212$.

Acknowledgment and provenance

The bounded-prime-gap application is motivated by Axiom Math's mathematical argument [1]. This manuscript records an independent Lean formalization of the specified results, with theorem names and commit identifiers to support reproducible review. Substantive exposition and attribution will be expanded in a subsequent draft.

References

[1] Axiom Math, *A New Bound for Small Gaps Between Primes*, preliminary draft, 3 September 2026. Project page: https://primegaps.axiommath.ai/.

[2] Reading Point, *lean-bgp212*: Lean source, verification records, and certificates. https://github.com/thinkthoughts/lean-bgp212.